ADC and DAC INTERFACING WITH 8051

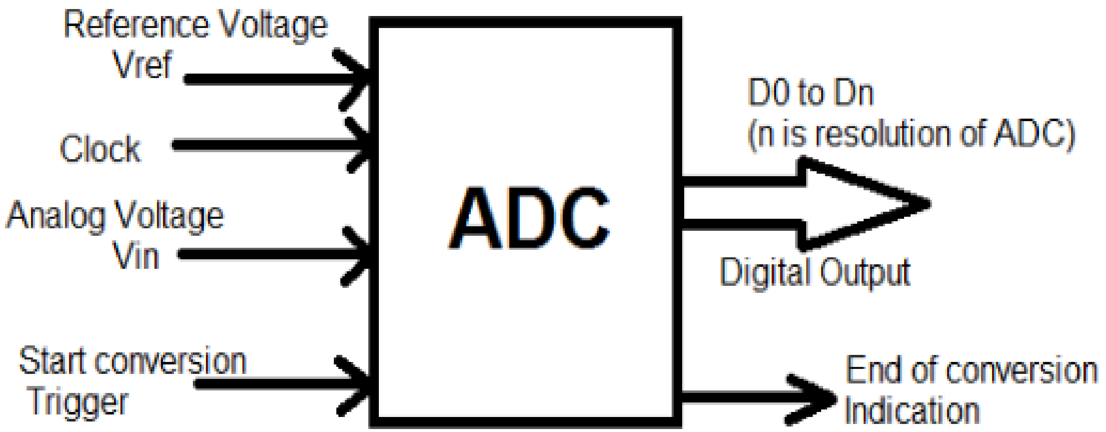
- Most of the physical quantity like temperature, displacement, pressure etc., all these are present in analog form. All these analog signals are processed and stored in memory is very complicated process.

- The analog sensors output data in an **analog format** which a **microcontroller cannot understand**. Therefore, to convert this analog data to a digital format, Analog to Digital converters or ADCs are used.

- An analog signal has a continuously changing amplitude with respect to time. A digital signal, on the contrary, is a stream of 0s and 1s.
- The digital-to-analog converter (DAC) is a device widely used to convert digital pulses to analog signals
- Audio and Video Devices: DACs are fundamental to audio and video devices, including smartphones, digital televisions, and audio players. They convert digital audio and video data into analog signals that our eyes and ears can perceive

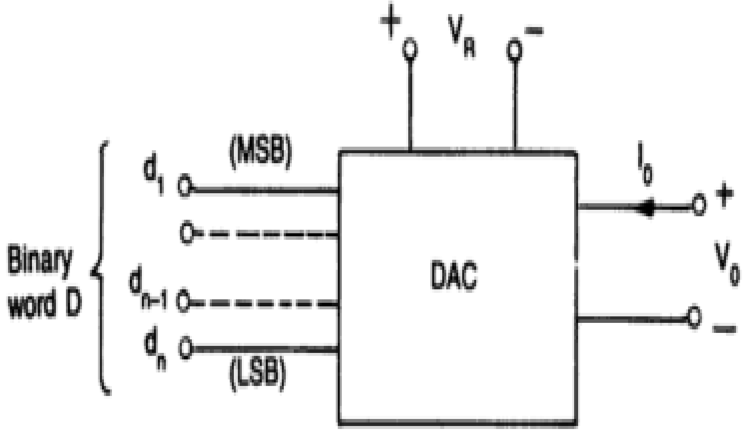
In general, there **five different types of ADCs** packaged as integrated circuits for use in electronic systems:

- Sigma-delta
- Successive approximation (SAR)
- Pipelined
- Dual-slope
- Flash



There are various ways of implementing DAC

- Weighted-Resistor DAC
- R-2R ladder DAC
- PWM type DAC



Characteristics / Specifications of a D/A converter

1. **Resolution:** The number of possible output levels the DAC is designed to reproduce. This is determined by the number of bits in the input binary word.
   A converter with 8 binary Inputs has 256 possible output levels, so its resolution is 1 part in 256. As another example, a 12-bit converter has a resolution of 1 part in 4096. Resolution is sometimes expressed as a percentage. The resolution of an 8-bit converter expressed as a percentage is (1/256) × 100 percent or about 0.39 percent.

Resolution describes the smallest possible change in the analog output voltage. The resolution should be as high as possible.

Its value depends on the number of bits in the digital input applied to the DAC. The higher the number of bits, the higher the resolution.

Resolution is defined as the ratio of change in analog output voltage resulting from a change of 1 LSB at the digital input.

$$Resolution = 2^n \quad \text{Or} \quad Resolution = \frac{V_{FS}}{2^n - 1}$$

where, n= Number of digital inputs

2. The **full-scale analog output voltage** V_{FS} is defined as the output voltage corresponding to the digital input with all digits 1.

3. The **accuracy** of D to A converter is a measure of the difference in between the expected output voltage and the actual output voltage.

   Accuracy is always specified n^th terms of the percentage of the full-scale output which means maximum output voltage. e.g. if the full-scale output s 15V and accuracy is ±0.1 percent then a maximum error is given by 0.001 x 15 = 0.015V or 15 mV

4. Linearity: The relation between the digital input and analog output should be linear. However, practically it is not so due to the error in the values of the resistor used for the resistive network.

5. Settling Time: Theoretically, the analog output voltage should change instantaneously in response to the change in its digital input.
Practically the analog output of the Digital to Analog converter does not change instantaneously. Due to the resistor and Op-Amp in the circuits, oscillations are observed at the output.
The time required to settle the analog output within 1/2 LSB of the final value after the change in digital input is called settling time. **The settling time should be as short as possible.**

6. Speed: It is defined as the time needed to perform a conversion from digital to analog. It is also defined as the **number of conversions that can be performed per second. The speed of DAC should be as high as possible.**

In an Analog-to-Digital Converter (**ADC**), **monotonicity** is a property of the transfer function that ensures the digital output code increases or decreases consistently in response to the analog voltage input increasing or decreasing consistently. This means that the curve of the input/output transfer function never reverses direction, even if it may be imperfect

A DAC is said to be monotonic if each digital-code increment produces an output equal to or larger than that of the preceding code. A DAC is generally expected to be monotonic to increments as small as an LSB, but in general the smallest increment for which the DAC remains monotonic determines its monotonicity.

Characteristics of ADC (Analog to Digital converter)

1. ***Resolution:*** Resolution is defined as the maximum number of digital output codes. This is the same as that of a DAC.

2. ***Conversion Time:*** It is the total time required to convert the analog signal into a corresponding digital output.

As we know, the conversion time depends on the conversion technique used for an ADC. The conversion time also depends on the propagation delay introduced by the circuit components. Conversion time should ideally be zero and practically as small as possible.

3. ***Quantization error*** is the difference between the analog signal and the closest available digital value at each sampling instant from the A/D converter.

4. The ***dynamic range*** refers to the ratio between the largest and smallest values an ADC can accurately measure.

5. ADC ***conversion speed*** refers to samples-per-second, and measures how quickly the device can accurately convert an analog signal/voltage

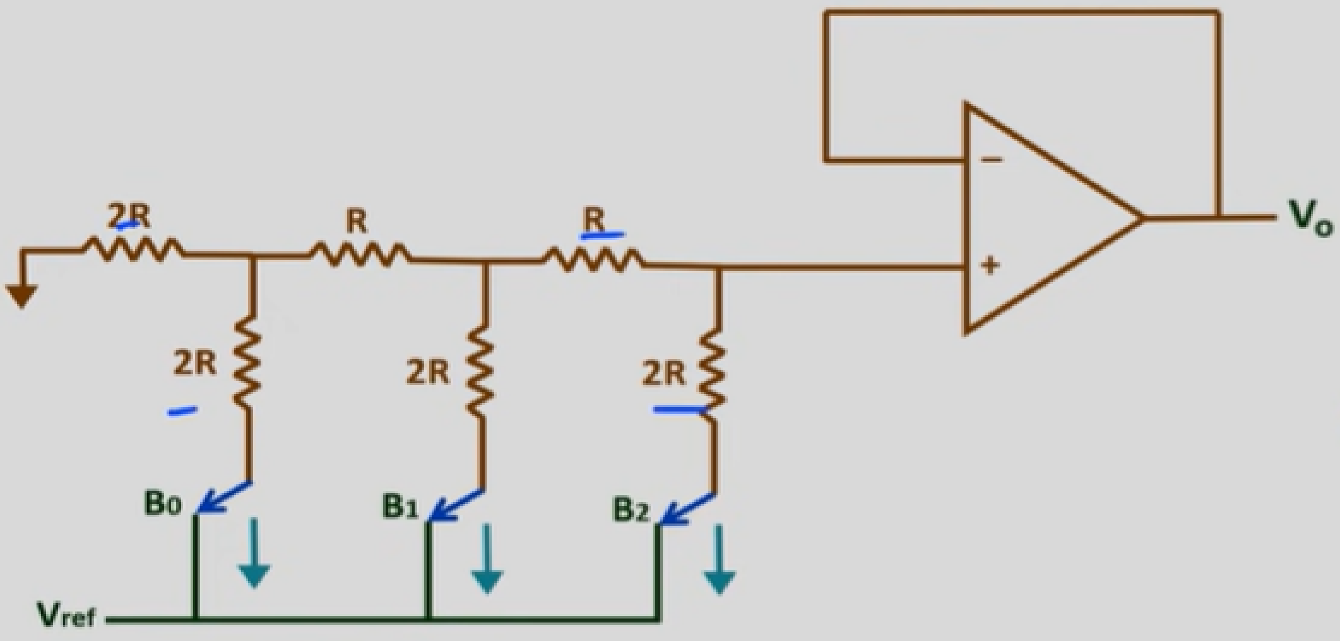
Microcontroller are used in wide variety of applications like for measuring and control of physical quantity like temperature, pressure, speed, distance, etc.

•In these systems microcontroller generates output which is in digital form but the controlling system requires analog signal as they don't accept digital data thus making it necessary to use DAC which converts digital data into equivalent analog voltage.

•In the figure shown, we use 8-bit DAC 0808. This IC converts digital data into equivalent analog Current. Hence we require an I to V converter to convert this current into equivalent voltage.

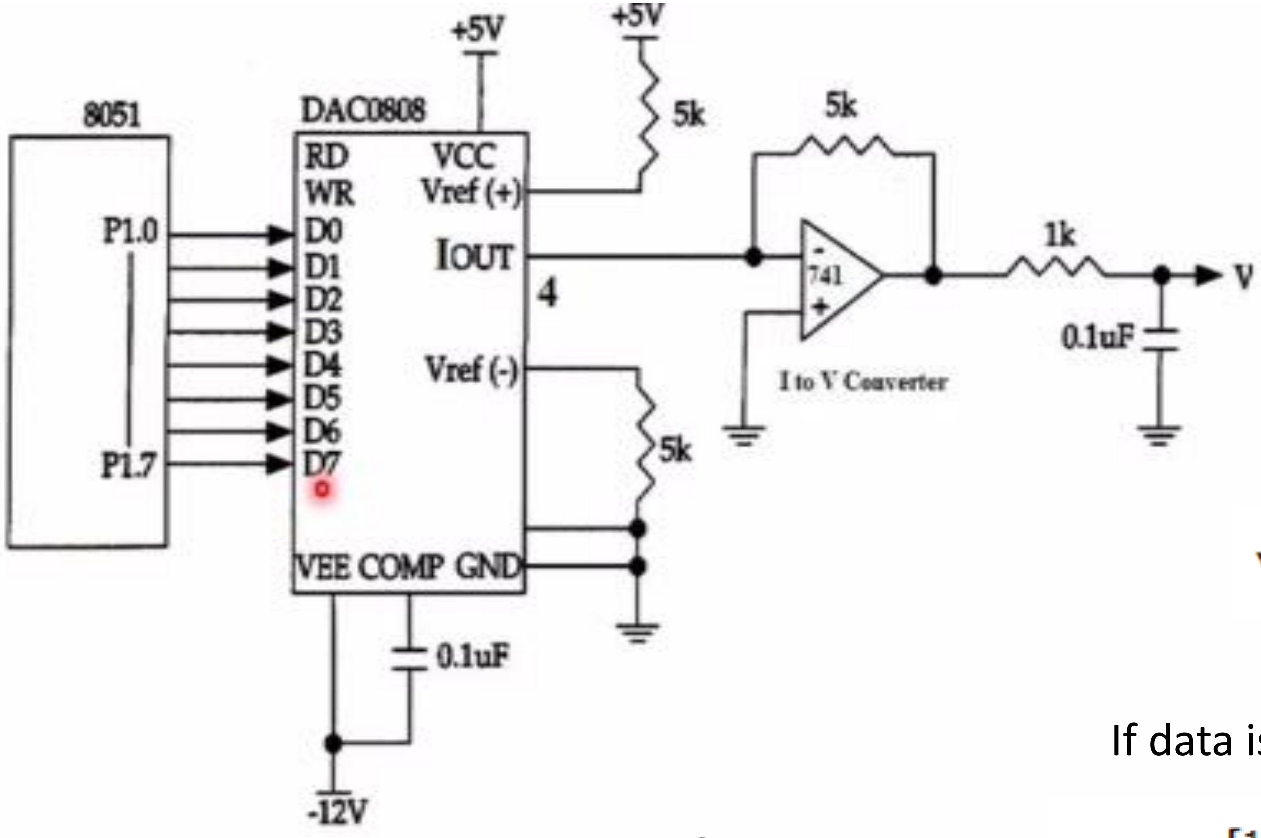
•According to theory of DAC Equivalent analog output is given as:

R-2R ladder DAC



$$V_o = V_{ref} \sum_{i=0}^{i=N-1} \frac{B_i}{2^{(N-i)}}$$

$$V_{ref}\left(\frac{B0}{2^{n}} + \frac{B1}{2^{n-1}} + \frac{B2}{2^{n-2}} + \text{-----} + \frac{B\,n-2}{2^{2}} + \frac{B\,n-1}{2^{1}}\right)$$



If data =00H [00000000], Vref= 10V

$$V0= 10\left[\frac{0}{2}+\frac{0}{4}+\frac{0}{8}+\frac{0}{16}+\frac{0}{32}+\frac{0}{64}+\frac{0}{128}+\frac{0}{256}\right]$$

Therefore, V0= 0 Volts

If data is 80H [10000000], Vref= 10V

$$V0=10\left[\frac{1}{2}+\frac{0}{4}+\frac{0}{8}+\frac{0}{16}+\frac{0}{32}+\frac{0}{64}+\frac{0}{128}+\frac{0}{256}\right]$$

Therefore, V0= 5 Volts

$$I_{out} = I_{ref}\left(\frac{D7}{2}+\frac{D6}{4}+\frac{D5}{8}+\frac{D4}{16}+\frac{D3}{32}+\frac{D2}{64}+\frac{D1}{128}+\frac{D0}{256}\right)$$

Different Analog output voltages for different Digital signal is given as

DATA	OUTPUT VOLTAGE
00H	0V
80H	5V
FFH	10V

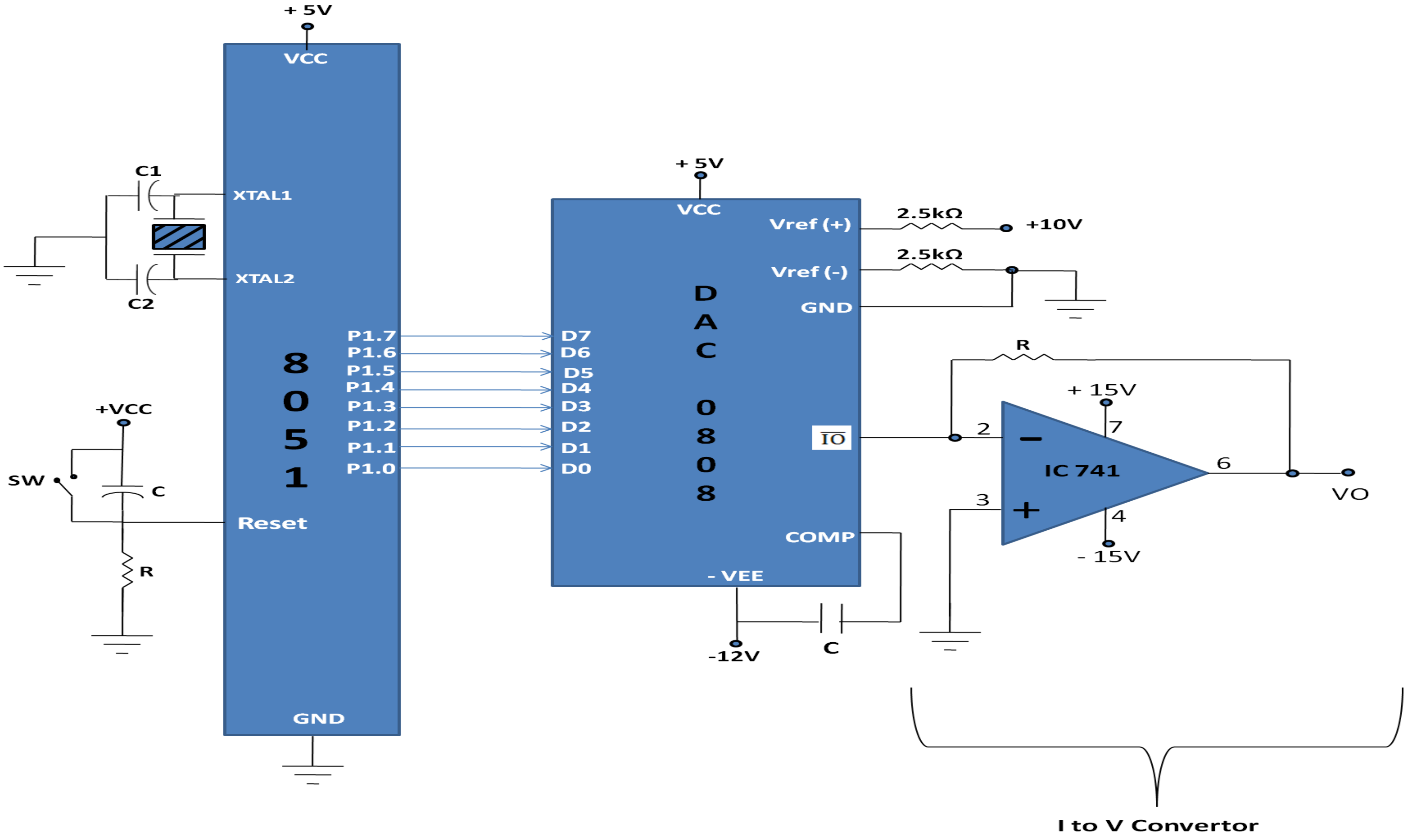
Assuming that R = 5K and I_{ref} = 2 mA, calculate V_{out} for the following binary inputs:

(a) 10011001 binary (99H) (b) 11001000 (C8H)

Solution:

(a) I_{out} = 2 mA (153/256) = 1.195 mA and V_{out} = 1.195 mA × 5K = 5.975 V

(b) I_{out} = 2 mA (200/256) = 1.562 mA and V_{out} = 1.562 mA × 5K = 7.8125 V


+ 5V
VCC
C1
XTAL1
XTAL2
C2
8051
P1.7
P1.6
P1.5
P1.4
P1.3
P1.2
P1.1
P1.0
+VCC
SW
C
Reset
R
GND
+ 5V
VCC
DAC 0808
Vref (+)
Vref (-)
GND
2.5kΩ
+10V
2.5kΩ
D7
D6
D5
D4
D3
D2
D1
D0
IO
COMP
- VEE
-12V
C
R
+ 15V
2
7
IC 741
6
3
4
- 15V
VO
I to V Convertor

Converting Iout to voltage in DAC0808:

➢ Ideally we connect the output pin lout, to a resistor, convert this current to voltage, and monitor the output on the scope.

➢ In real life, however, this can cause inaccuracy since the input resistance of the load where it is connected will also affect the output voltage.

➢ For this reason, the lref current output is isolated by connecting it to an op-amp such as the 741 with Rf = 5K ohms for the feedback resistor.

➢ Assuming that R= 5K ohms, by changing the binary input, the output voltage changes as shown in Example 2.

Example 2: Inorder to generate a stair-step ramp, set up the circuit in figure and connect the output to an oscilloscope.

Then write a program to send data to the DAC to generate a stair-step ramp.

Solution:

```
        CLR A
AGAIN:  MOV P1,A ; SEND DATA TO DAC
        INC A ; COUNT FROM 0 TO FFH
        ACALL DELAY ; LET DAC RECOVER
        SJMP AGAIN
```

Thus Microcontroller 8051, when interfaced with DAC-0808 and at the output we get Square wave and Triangular wave respectively for a given set of digital inputs.

http://what-when-how.com/8051-microcontroller/dac-interfacing/

https://www.tutorialspoint.com/interfacing-dac-with-8051-microcontroller

ANALOG-TO-DIGITAL CONVERTER (ADC) INTERFACING:

- An input signal which varies from 0 to 8 volt,
- use a 3-bit ADC to convert this signal to binary data.
- A 3-bit ADC can represent 2^3 or 8 different voltage levels using 3 bits of data.

Input voltage	Binary equivalent
0-1 volt	000B
1-2 volt	001B
2-3volt	010B
3-4 volt	011B
4-5 volt	100B
5-6 volt	101B
6-7 volt	110B
7-8 volt	111B

- The tiniest change we can detect is that of 1 volt. If the change is smaller than 1 volt, the ADC can’t detect it. This minimum change that an ADC can detect is known as the step size of the ADC.

- Step size=Vmax-Vmin/2n

(where n is the number of bits(resolution) of an ADC)

Input voltage	Binary equivalent
0-1 volt	000B
1-2 volt	001B
2-3volt	010B
3-4 volt	011B
4-5 volt	100B
5-6 volt	101B
6-7 volt	110B
7-8 volt	111B

- An ADC has n-bit resolution where n can be 8, 10, 12, 16 or even 24 bits.
- The higher-resolution ADC provides a smaller step size, where step size is the smallest change that can be discerned by an ADC. This is shown in table 5.3

Table 5.3 Resolution Vs Step Size for ADC

n-bit	Number of steps	Step Size (mV)
8	256	5/256 = 19.53
10	1024	5/1024 = 4.88
12	4096	5/4096 = 1.2
16	65536	5/65536 = 0.076

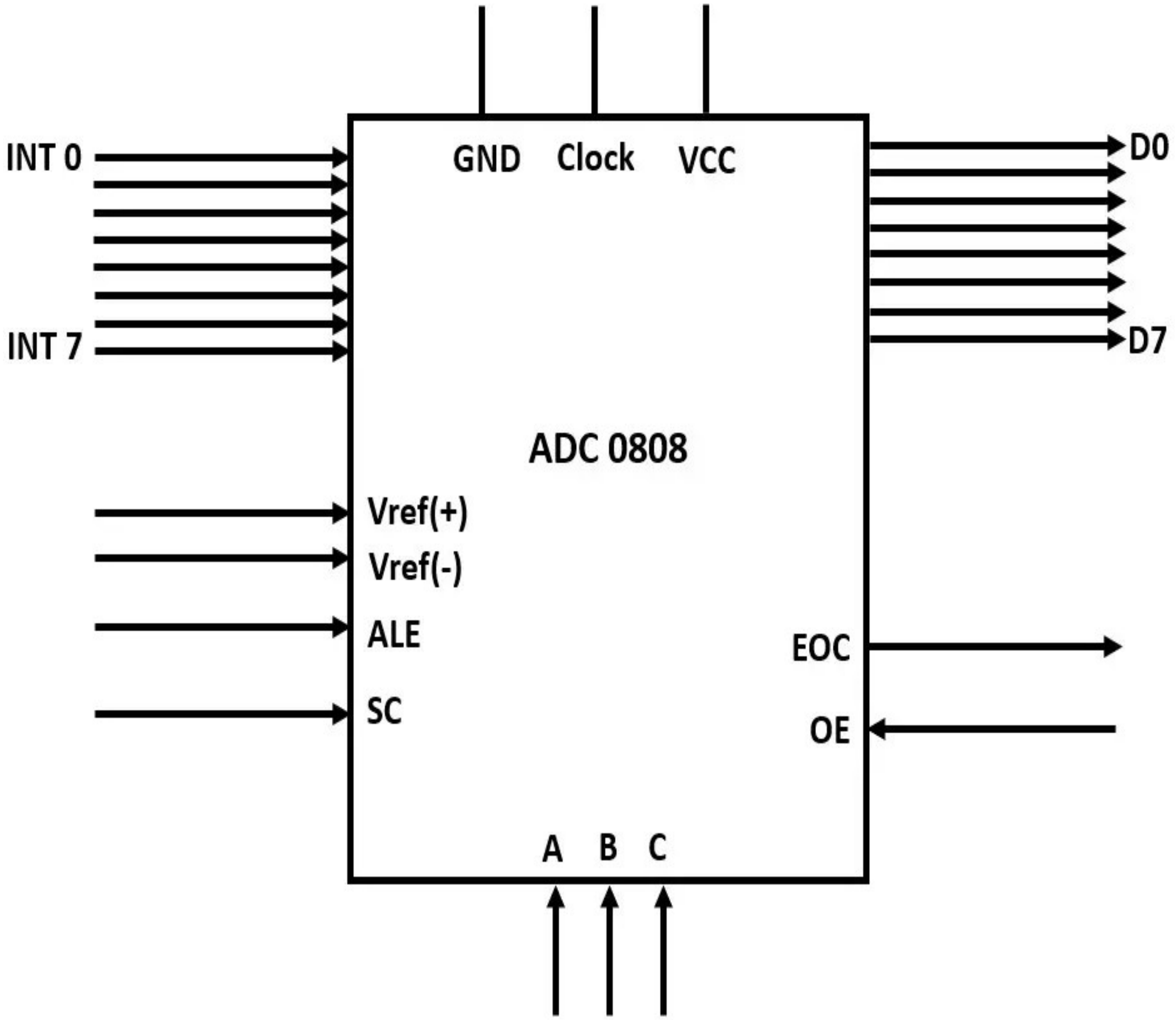
Notes: $V_{CC} = 5$ V

Step size (resolution) is the smallest change that can be discerned by an ADC.

- In addition to resolution, conversion time is another major factor in judging an ADC.
- Conversion time is defined as the time it takes the ADC to convert the analog input to a digital (binary) number.
- The ADC chips are either parallel or serial.
- In parallel ADC, we have 8 of more pins dedicated to bringing out the binary data, but in serial ADC we have only one pin for data out.

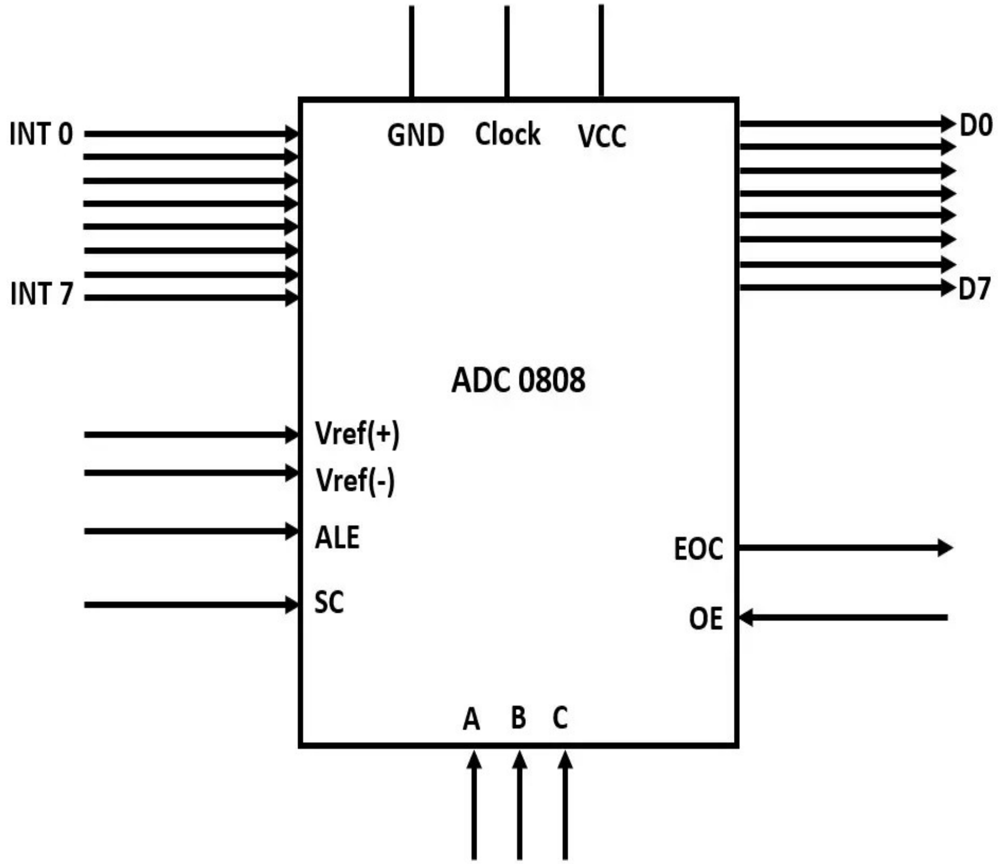
ADC 0808

- The ADC 0808 is a popular 8-bit ADC with a step size of 19.53 millivolts.
- It does not have an internal clock. Therefore, it requires a clock signal from an external source.
- It has eight input pins, but only one of them can be selected at a time because it has eight digital output pins.
- It uses the principle of **successive approximation** for calculating digital values, which is very accurate for performing 8-bit analog to digital conversions.



Input pins (INT0-INT7)

- The ADC 0808 has eight input analog pins. These pins are multiplexed together, and only one of them can be selected using three select lines.



Select lines and ALE

- It has three select lines, namely A, B, and C, that are used to select the desired input lines. The ALE pin also needs to be activated by a low to high pulse to select a particular input. The input lines are selected as follows:

ADC interface

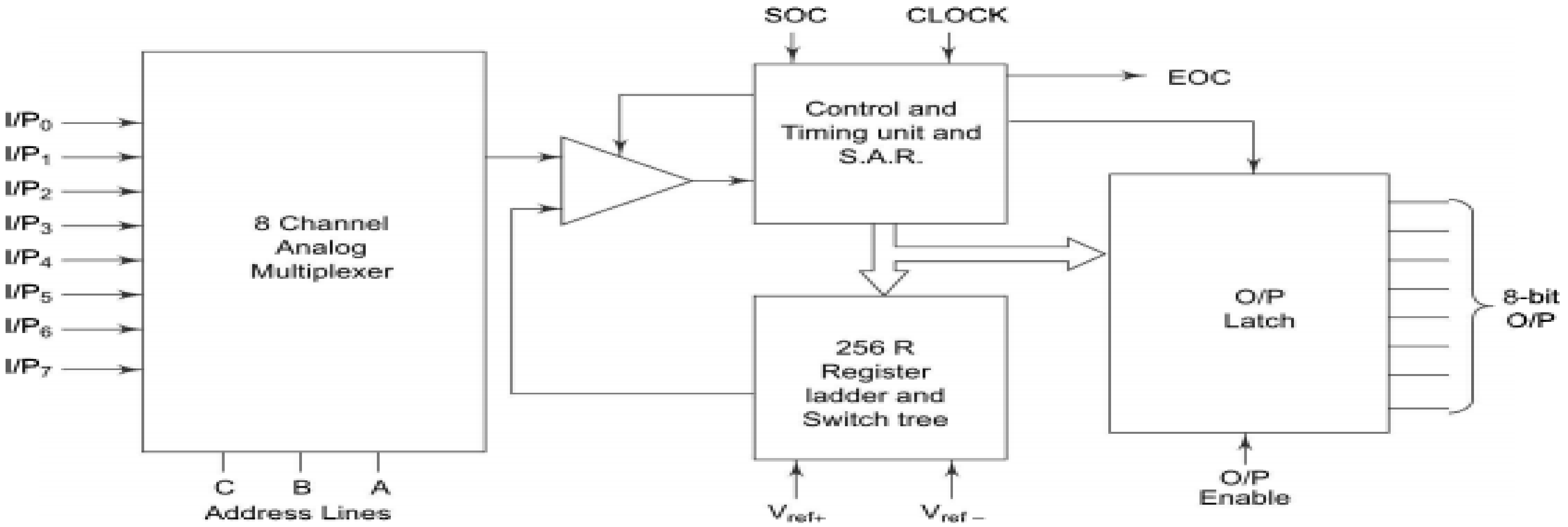


Fig. 5.37(a) *Block Diagram of ADC 0808/0809*

Whatever may be the technique used for conversion, a general algorithm for ADC interfacing contains the following steps.

1. Ensure the stability of analog input, applied to the ADC
2. Issue start of conversion (SOC) pulse to ADC
3. Read end of conversion (EOC) signal to mark the end of conversion process
4. Read digital data output of the ADC as equivalent digital output

Table 5.13

Analog I/P selected	*Address lines*		
	C	*B*	*A*
I/P 0	0	0	0
I/P 1	0	0	1
I/P 2	0	1	0
I/P 3	0	1	1
I/P 4	1	0	0
I/P 5	1	0	1
I/P 6	1	1	0
I/P 7	1	1	1

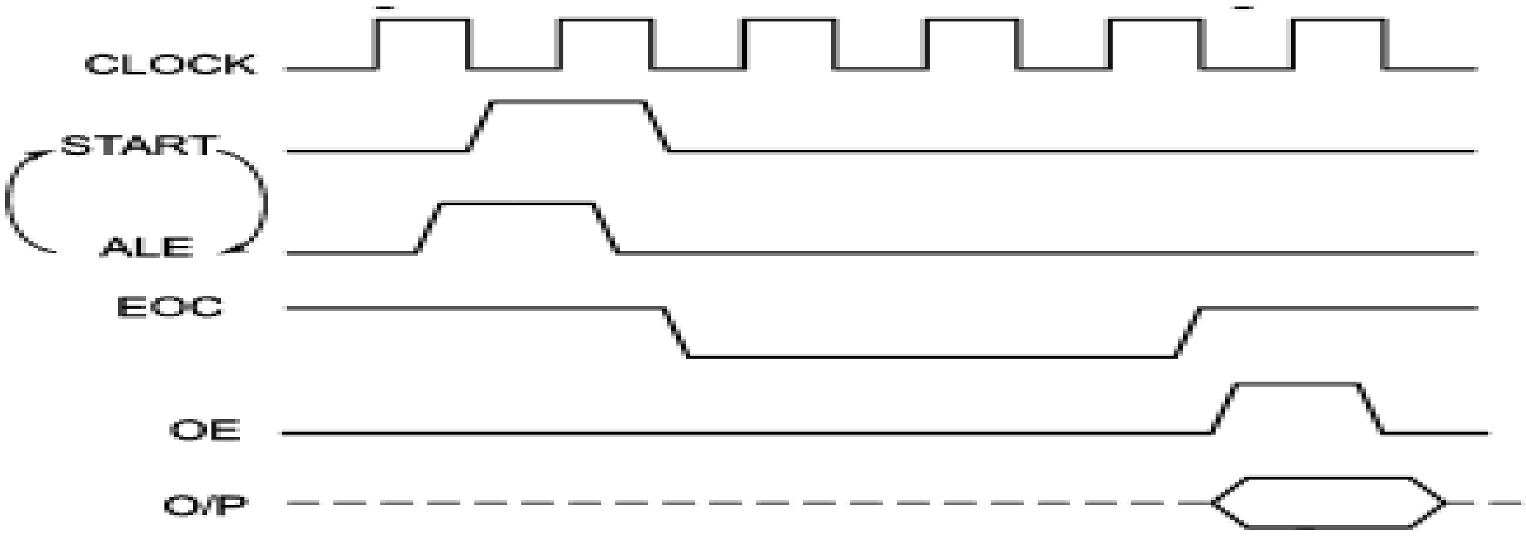


Fig. 5.38 *Timing Diagram of ADC 0808*

- **Steps to Be followed For Data Conversion:**
  - The following steps must be followed for data conversion by the ADC804 chip:
    - Make CS= 0 and send a L-to-H pulse to pin WR to start conversion.
    - Monitor the INTR pin, if high keep polling but if low, conversion is complete, go to next step.
    - Make CS= 0 and send a H-to-L pulse to pin RD to get the data out.
  - Figure 5.6 shows the timing diagram for ADC process.

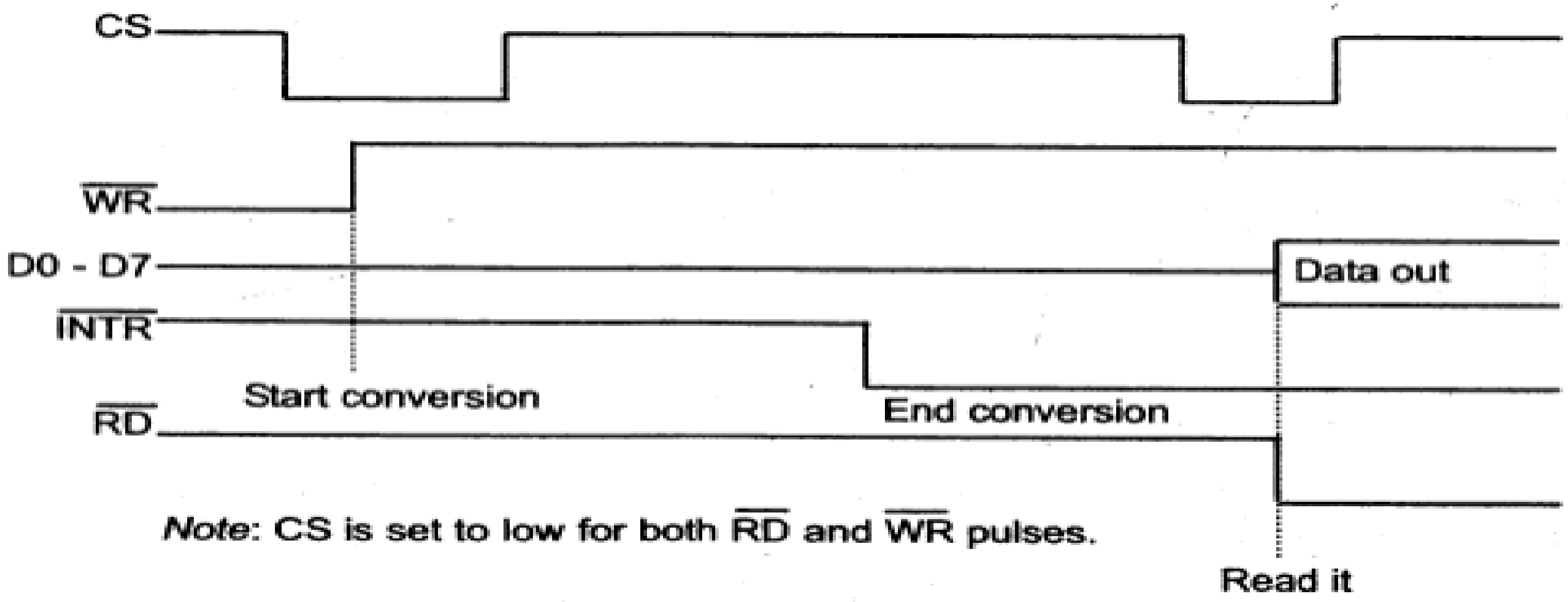
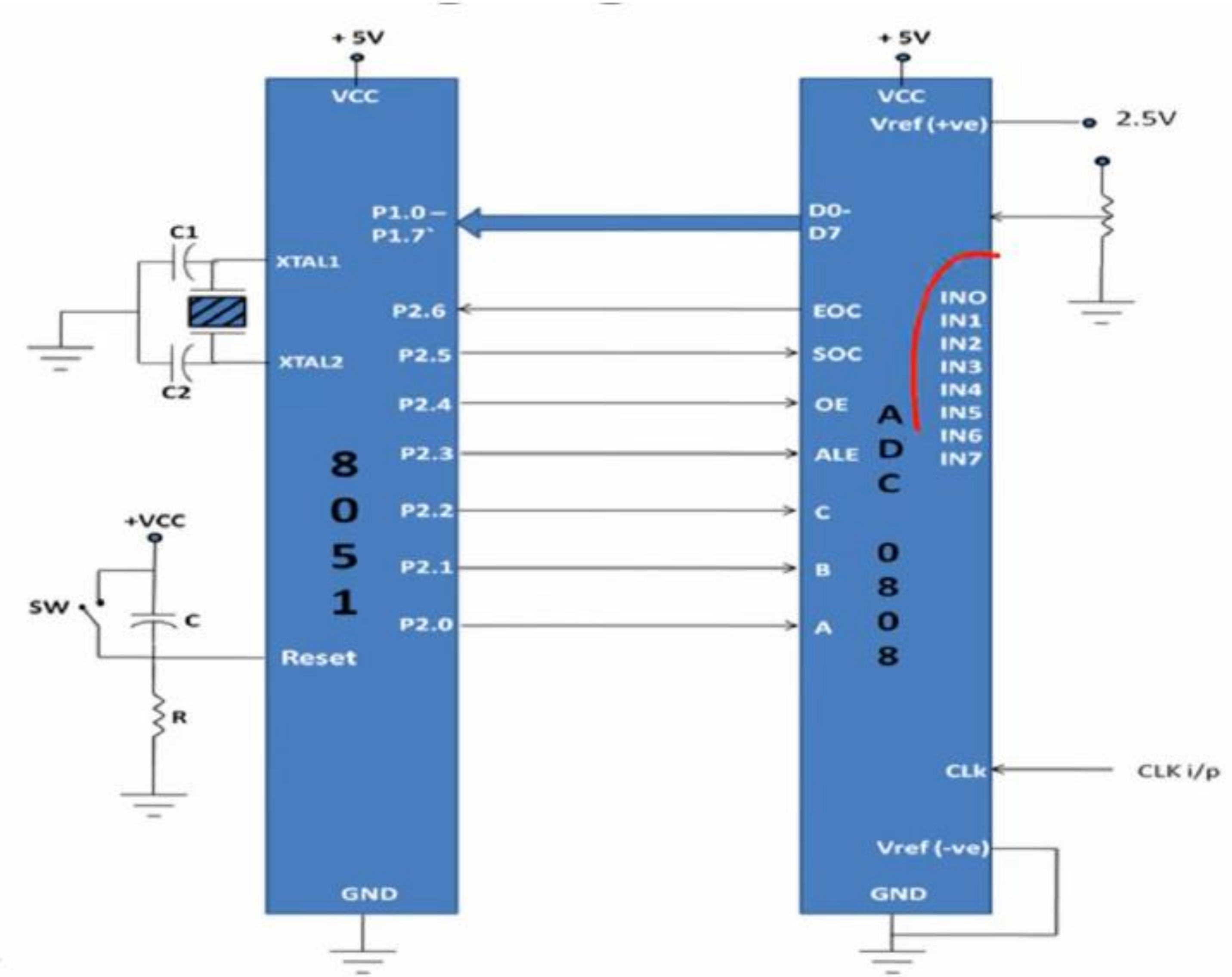


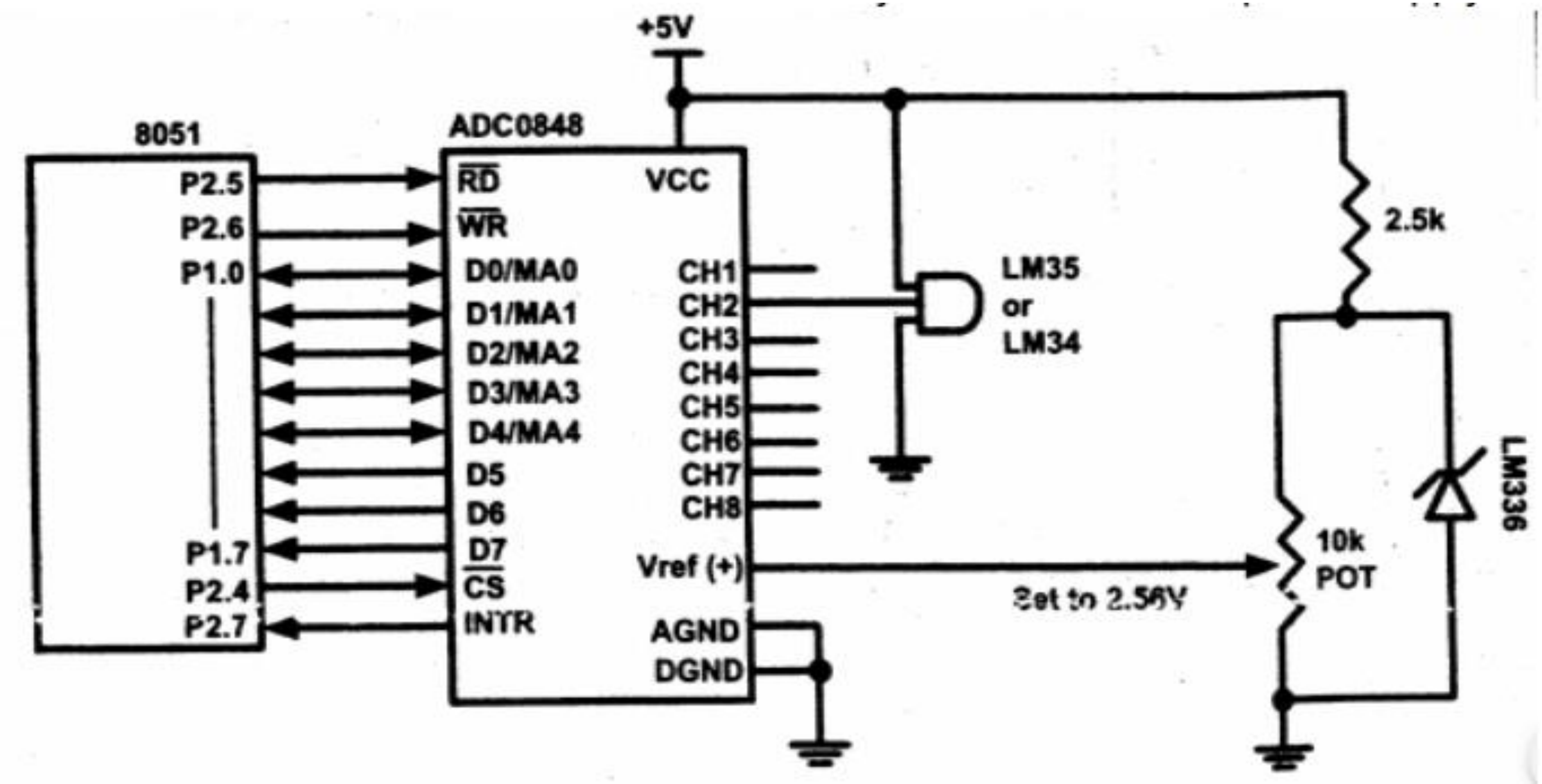
Fig : Read and Write Timing for ADC0804



1. Apply the analog input to one of the input channel.
2. Select the respective channel
3. Send output High pulse on ALE
4. Send output High pulse on SoC
5. Wait for low pulse on EoC
6. Make OE high to enable microcontroller to read output data
7. Read the digital output on port 1
8. Repeat steps 3 to 7 for next conversion.

https://embetronicx.com/tutorials/microcontrollers/8051/adc0804-interfacing-with-8051/

https://circuitdigest.com/microcontroller-projects/interfacing-adc0808-with-8051-microcontroller



8051 Connection to ADC0848 and Temperature sensor

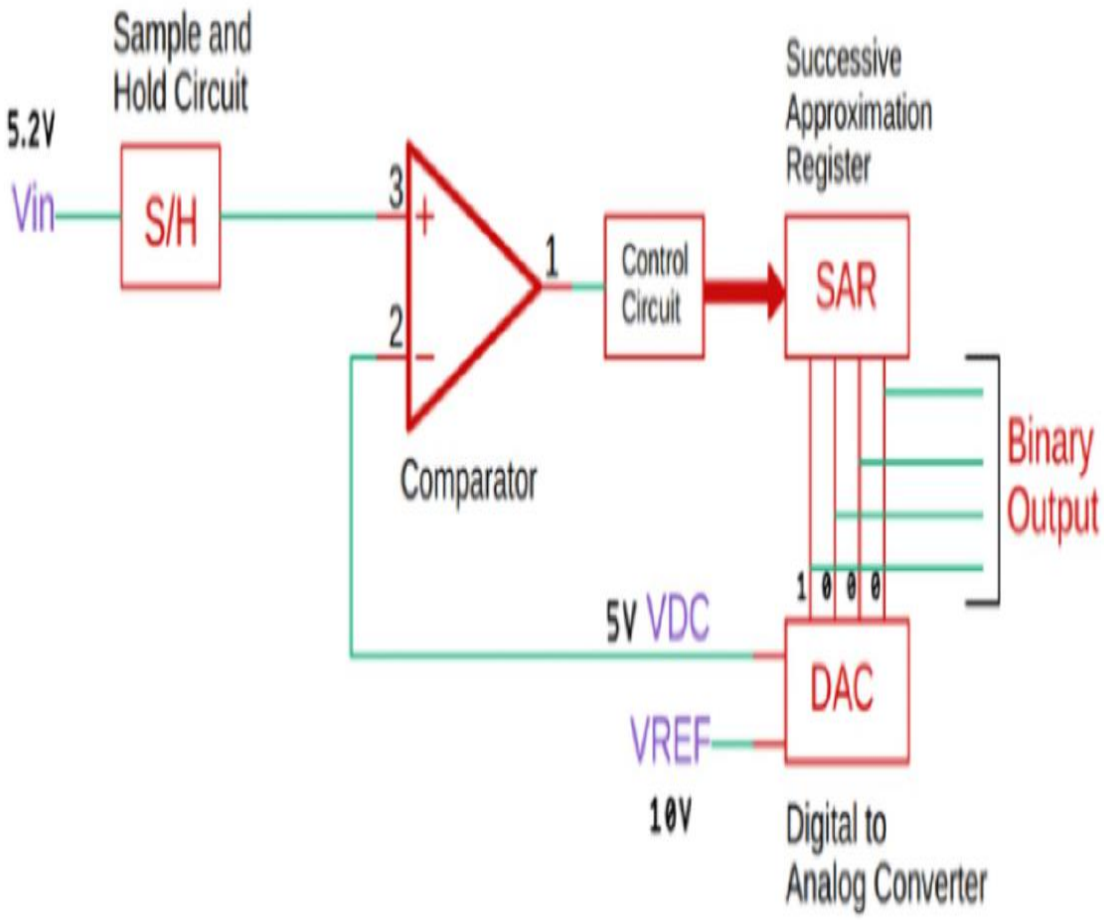
SUCCESSIVE APPROXIMATION ADC

A successive-approximation ADC is a type of analog-to-digital converter (ADC) that converts **a continuous analog waveform into a discrete digital representation** using a binary search through all possible quantization levels before finally converging upon a digital output for each conversion

As you can see, this ADC consists of a **comparator, a digital to analog converter, and a successive approximation register** along with the control circuit,

Now whenever a new conversion starts, the **sample and hold circuit samples the input signal.** And that signal is compared with the specific output signal of the DAC.

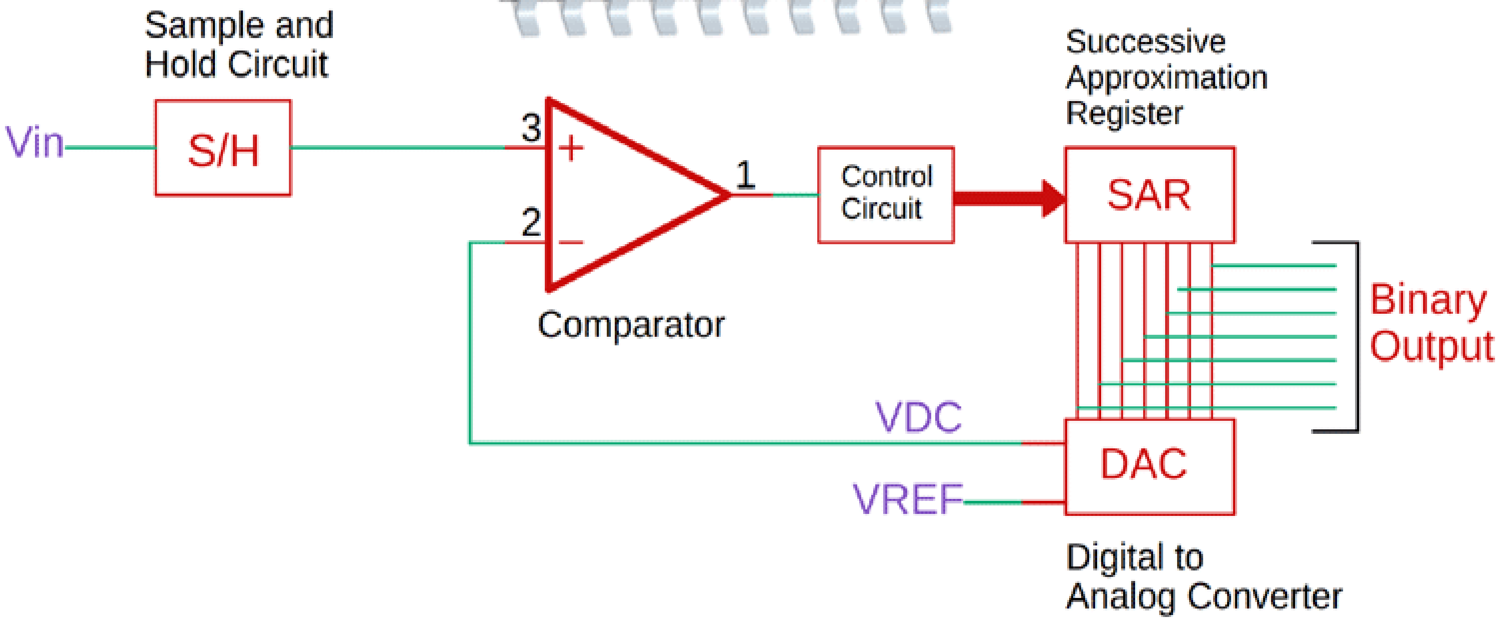
Samples and holds analog signal For **each bit**, a **successive approximation register analyzes** it and outputs binary to a digital-to-analog **converter that is dependent on the current bit** and all **previously approximated** After the DAC, a **comparator compares the voltage of the analog to the digital**, and **continually loops** until they are equivalent

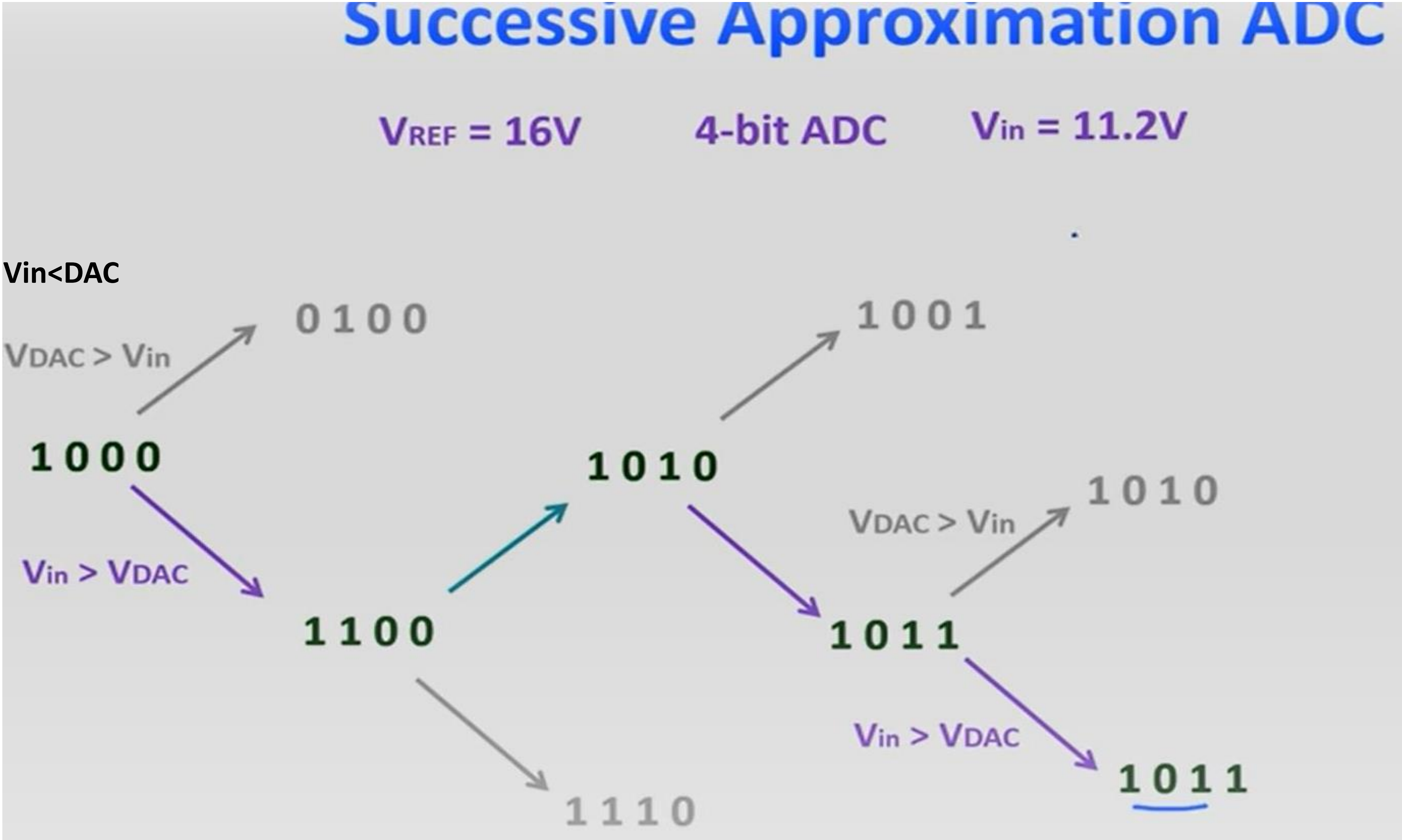


Vin > Vdc = SET YOUR NEXT BIT

Vin<Vdc = SET BIT IS GOING TO RESET AND WILL SET NEXT BIT

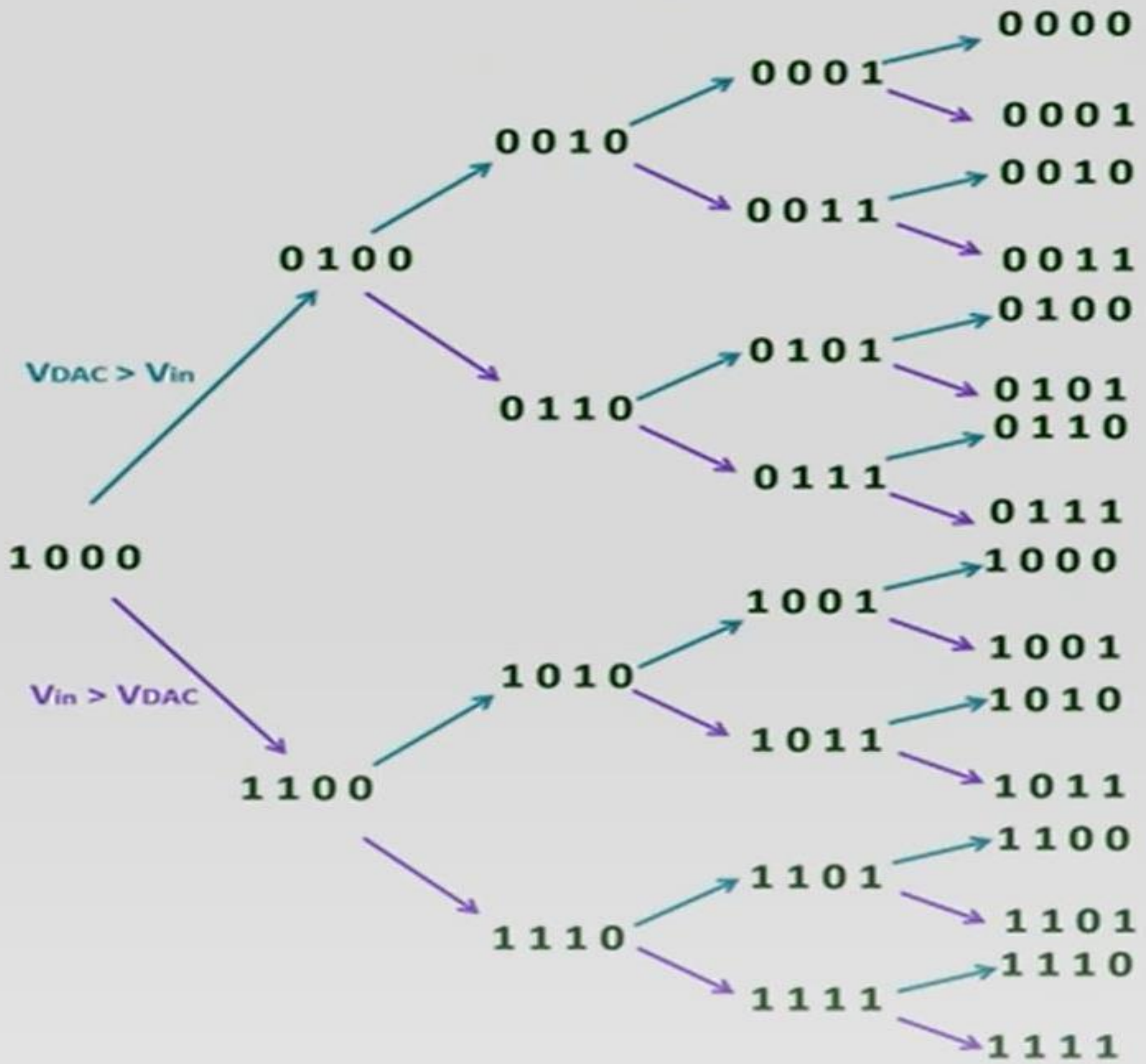
SUCCESSIVE APPROXIMATION ADC





If Vin>DAC voltage ,then Keep the bit and change next bit to 1
If Vin<DAC, Make the bit 0, change the next bit to 1

Successive Approximation ADC



Advantages:

Conversion time is very small and independent of the amplitude of the signal.
Good speed to power ratio

Disadvantages:

It may be fast, but it’s slower than a flash ADC, that uses a linear ‘voltage ladder’ and compares the input voltage to each ‘rung’ of the ladder
It’s a somewhat complex circuit